

CS471 - Web Technologies (Laboratory)

Lab 3: Django Views, URLs, and Templates

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GitHub Repository: github.com/lujain157/CS471-Labs

Overview

This lab practices Django views, URL routing, URL and query parameters, templates, template variables, and dynamic display of book information.

Task 1 — Basic View and Query Parameter

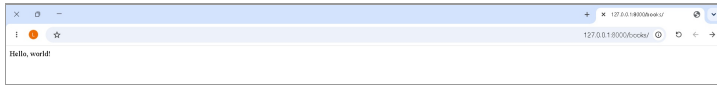


Figure 1: Default output: "Hello, world!"

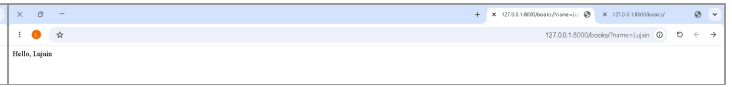


Figure 2: Output with ?name=Lujain query parameter

The view reads the `name` query parameter and shows a default value when no name is supplied.

Task 2 — URL Integer Parameter

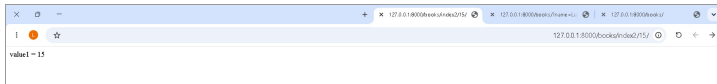


Figure 3: index2/15/ returns value1 = 15

An integer value is passed through the URL path to the Django view.

Task 3 — Book Module Template

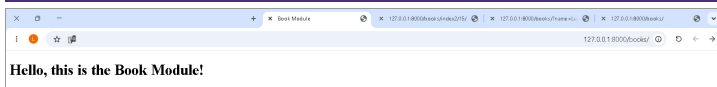


Figure 4: Book Module rendered via HTML template

The Book Module view was rendered using an HTML template.

Task 4 — Template Variable

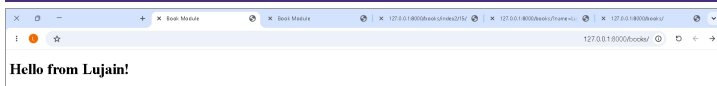


Figure 5: Template variable output: "Hello from Lujain!"

Data was passed from the Django view to the HTML template using a context variable.

Task 5 — User Module

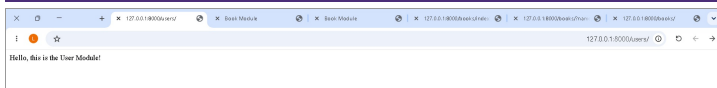


Figure 6: User Module accessed via its URL

The User Module was configured and accessed through its own URL route.

Task 6 — URL and Parameter Testing

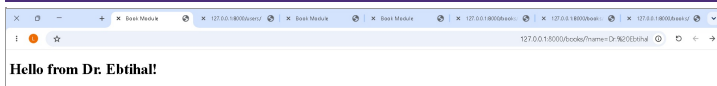


Figure 7: ?name=Dr. Ebtihal query parameter

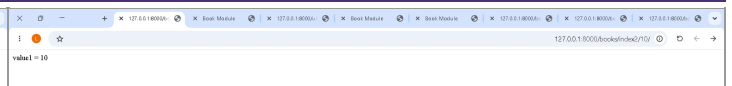


Figure 8: index2/10/ returns value1 = 10

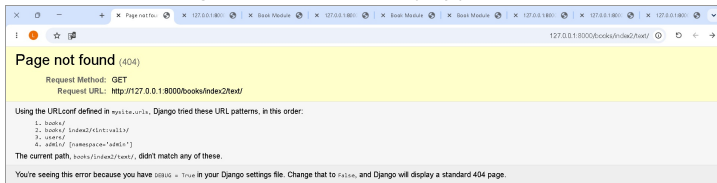


Figure 9: index2/text/ does not match <int:val1> — 404 Not Found

The URL pattern uses `<int:val1>`, so integer values are accepted while text input does not match the pattern and returns a 404 response.

Task 7 — Display Book Details

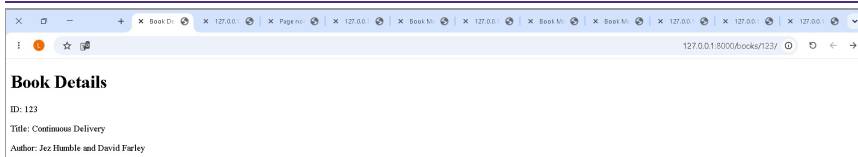
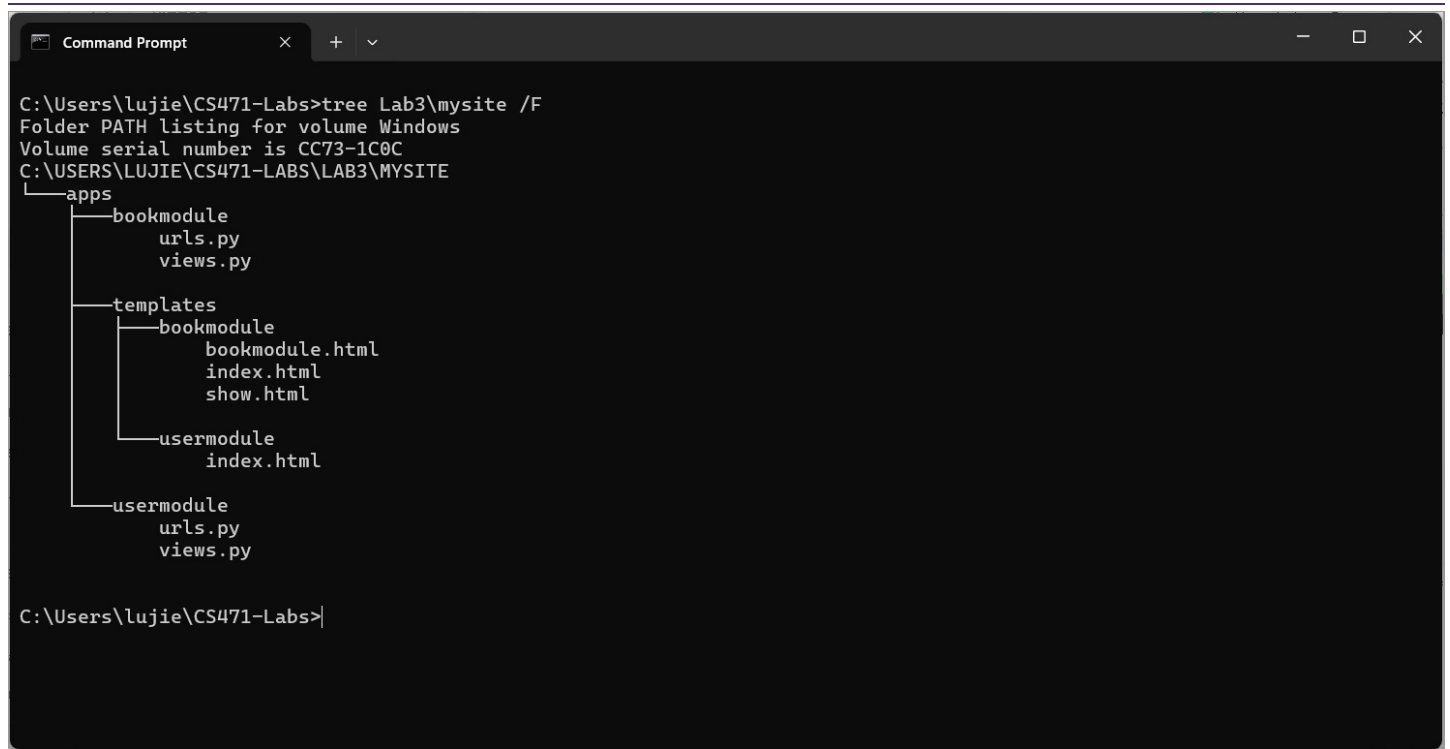


Figure 10: Book Details for ID 123 — Continuous Delivery by Jez Humble and David Farley

The book ID is received from the URL, the matching book is selected in the view, and its details are passed to `show.html` and displayed using Django template variables.

Project File Structure



```
Command Prompt
C:\Users\lujie\CS471-Labs>tree Lab3\mysite /F
Folder PATH listing for volume Windows
Volume serial number is CC73-1C0C
C:\USERS\LUJIE\CS471-LABS\LAB3\MYSITE
├── apps
│   ├── bookmodule
│   │   ├── urls.py
│   │   └── views.py
│   ├── templates
│   │   ├── bookmodule
│   │   │   ├── bookmodule.html
│   │   │   ├── index.html
│   │   │   └── show.html
│   │   └── usermodule
│   │       └── index.html
│   └── usermodule
│       ├── urls.py
│       └── views.py
└── usermodule
    ├── urls.py
    └── views.py

C:\Users\lujie\CS471-Labs>
```

Figure 11: Lab 3 project file structure (output of `tree Lab3\mysite /F`)

Steps Performed

- Created Django views and configured URL patterns for each task.
- Used query parameters and integer URL parameters (`<int:val1>`).
- Created HTML templates and passed data using context variables.
- Configured and tested the User Module route.
- Tested URL converters, confirming correct 404 handling for invalid input.
- Displayed book details dynamically based on the URL-provided book ID.

Conclusion

The lab successfully demonstrated Django views, URL routing, URL and query parameters, templates, context variables, and dynamic content rendering.